

ShellArc Railgun Architecture - Technical Note

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1 Introduction

This is a mathematical model designed for schedule and production management in a 2D animation production pipeline.

2 Definition of the Three-Dimensional Domain Used to Demonstrate Progress

Define a discrete three-dimensional lattice space Γ , in which each discrete point is defined as (x, y, z) , where

$$\begin{cases} x \in [1, c_1] \cap \mathbb{Z}_+ & (c_1 \in (1, \infty) \cap \mathbb{Z}_+) \\ y \in [0, F] \cap \mathbb{Z} & (F \in (1, \infty) \cap \mathbb{Z}_+) \\ z \in [0, c_2] \cap \mathbb{Z}_+ & (c_2 \in (1, \infty) \cap \mathbb{Z}_+) \end{cases}$$

Here c_1 , F , and c_2 denote the total cut count, the component-division fineness, and the project total length (day count), respectively.

Submissions are represented by a group of points defined as

$$\begin{cases} M = \{ p_k = (x_k, y_k, z_k) \mid k = 1, 2, \dots, |M| \} \cap \Gamma \\ \forall (x, y) \in \mathbb{Z}^2, \exists! z \in \mathbb{Z} \text{ s.t. } (x, y, z) \in M \end{cases}$$

3 Evaluating Overall Progress and Status

In Γ , define face I , which represents an ideal project progress, as

$$I : \begin{cases} z = (1 - t) \sqrt{x^2 + y^2} \\ t = \frac{c_2^2}{\sqrt{c_1^2 + 100}} \in \mathbb{R} \quad (t = \text{const.}) \end{cases}$$

Relocate the points in M and define the resulting group as G , illustrated by the map $f : M \rightarrow G$.

Define $f(p_k) = g_k = (x'_k, y'_k, z'_k)$, where

$$\begin{cases} z'_k = z_k \\ \forall i, j \in \{1, 2, \dots, |M|\}, \quad i \neq j \implies f(p_i) \neq f(p_j) \\ \forall (x, y, z) \in G, \quad \mathcal{N}(x, y, z) \cap (G \setminus \{(x, y, z)\}) \neq \emptyset \quad \dots(*) \end{cases}$$

where the neighbourhood is defined as

$$\mathcal{N}(x, y, z) = \{(x \pm 1, y, z), (x, y \pm 1, z)\}.$$

Define

$$E(x, y, z_k) = \sqrt{x^2 + y^2} - \epsilon z_k, \quad \epsilon = \frac{1}{1-t} = \text{const.}$$

M is relocated to form G such that, for $k \in \{1, 2, \dots, |M|\}$,

$$g_k = \arg \min_{g \in \mathcal{A}_k} |E(x, y, z_k)|,$$

where $\mathcal{A}_k$ is the set of dynamically determined available positions for the k -th point, subject to constraint $(*)$.

Since $G \subseteq \Gamma$, let T denote today's date, and define

$$G' = G \cup \{(x, y, T)\} \quad \text{s.t.} \quad |G'| = |\Gamma|, \quad (x, y, T) \notin G.$$

Solve for the total offset R between I and the points in G' :

$$R = \sum_{p \in G'} |p_z - z_I(p)|, \quad \text{where } (p_x, p_y, z_I(p)) \in I.$$

The mean

$$\mu = \frac{R}{|\Gamma|}$$

illustrates the overall progress relative to the ideal progress, and

$$\sigma = \sqrt{\frac{1}{|\Gamma|} \sum_{p \in G'} \left(|p_z - z_I(p)| - \mu \right)^2}$$

illustrates the deviation of progress across the project.

4 Evaluating Momentum of Each Part of the Project

For cut number $x_k \in [1, C_1]$, the group of points $\{(x, y, z) \in M \mid x = x_k\}$ is interpolated by a cubic interpolation function S_c such that $y \approx S_c(z)$.

Define the partial derivative of y with respect to z at $x = x_k$ as

$$\left(\frac{\partial y}{\partial z} \right)_x \bigg|_{x=x_k} = \frac{d}{dz} S_c(z).$$

The partial derivative of y with respect to z at $x = x_k$ and component $y = y_k$ represents the momentum of the project at component y_k of cut x_k .

To capture the current momentum of a cut, the momentum between the day of the latest submitted component and today (day T) must be calculated and included in the analysis.

The extended point collection M' is defined as

$$M' = M \cup \left\{ (x_k, y_{\text{next}}, T) \mid x_k \in [1, C_1], \quad y_{\text{next}} = \left(\arg \max_{\substack{(x,y,z) \in M \\ x=x_k}} y \right) + 1 \right\}.$$

The momenta of component y_k of each cut x_k , given by $\left(\frac{\partial y}{\partial z} \right)_x \Big|_{x=x_k, y=y_k}$, are remapped and normalised as follows:

$$[\min(\mathcal{A}), \max(\mathcal{A})] \longrightarrow [0, 1], \quad \text{where } \mathcal{A} = \left\{ \left(\frac{\partial y}{\partial z} \right)_x \Big|_{x=x_k, y=y_k} \mid (x_k, y_k, z_k) \in M' \right\}.$$

The resulting float value represents the momentum of a specific component of a given cut.

Here, define

$$M_{\text{frontier}} = \left\{ (x_k, y_k, T) \mid x_k \in [1, C_1], \quad y_k = \arg \max_{\substack{(x,y,z) \in M \\ x=x_k}} y \right\} \subseteq M',$$

and

$$M'_{\text{frontier}} = \left\{ \forall p_i \in M_{\text{frontier}}, \quad p_i = (x_i, y_i) \mid x_1 < x_2 < \dots < x_{|M_{\text{frontier}}|} \right\}.$$

Define

$$\mathcal{F} = \bigcup_{i=1}^{|M'_{\text{frontier}}|-1} \left\{ (1-m)p_i + m p_{i+1} \mid m \in [0, 1], \quad p_i, p_{i+1} \in M'_{\text{frontier}} \right\}.$$

$\mathcal{F}$ illustrates the current progress of each cut throughout the project.